Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution Affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Algorithms Laboratory		
Academic Year	2025–2026 (Even)	Batch	2023–2027
Name	**Jayasimman J**	Register No.	**3122235001060**
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Experiment 3: Regression Analysis using Linear and Regularized Models

1 Aim and Objective

To implement linear and regularized regression models for predicting a continuous target variable, evaluate their performance using multiple metrics, visualize model behavior, and analyze overfitting, underfitting, and bias–variance characteristics.

2 Dataset Description

A real-world regression dataset containing numerical and categorical features related to loan applications is used. The target variable is the **loan amount sanctioned**.

Dataset Reference:

- Kaggle: Predict Loan Amount Dataset

3 Preprocessing Steps

- The dataset was loaded into a Pandas DataFrame, and features and target labels were separated.
- Missing values in numerical features were imputed using the median strategy with an indicator for missingness.
- Categorical features were imputed with a constant value ("Missing") and encoded using OneHotEncoder.
- Numerical features were standardized using StandardScaler to ensure zero mean and unit variance.
- A train–validation–test split was performed to avoid data leakage.

4 Implementation Details

- Implemented Linear Regression, Ridge, Lasso, and Elastic Net models.

- Hyperparameter tuning was performed using GridSearchCV.
- Models were evaluated using CV R^2, MAE, MSE, RMSE, and R^2.
- Visualizations include predicted vs actual plots, residual plots, learning curves, and coefficient comparisons.

5 Visualizations

5.1 Exploratory Data Analysis (EDA)

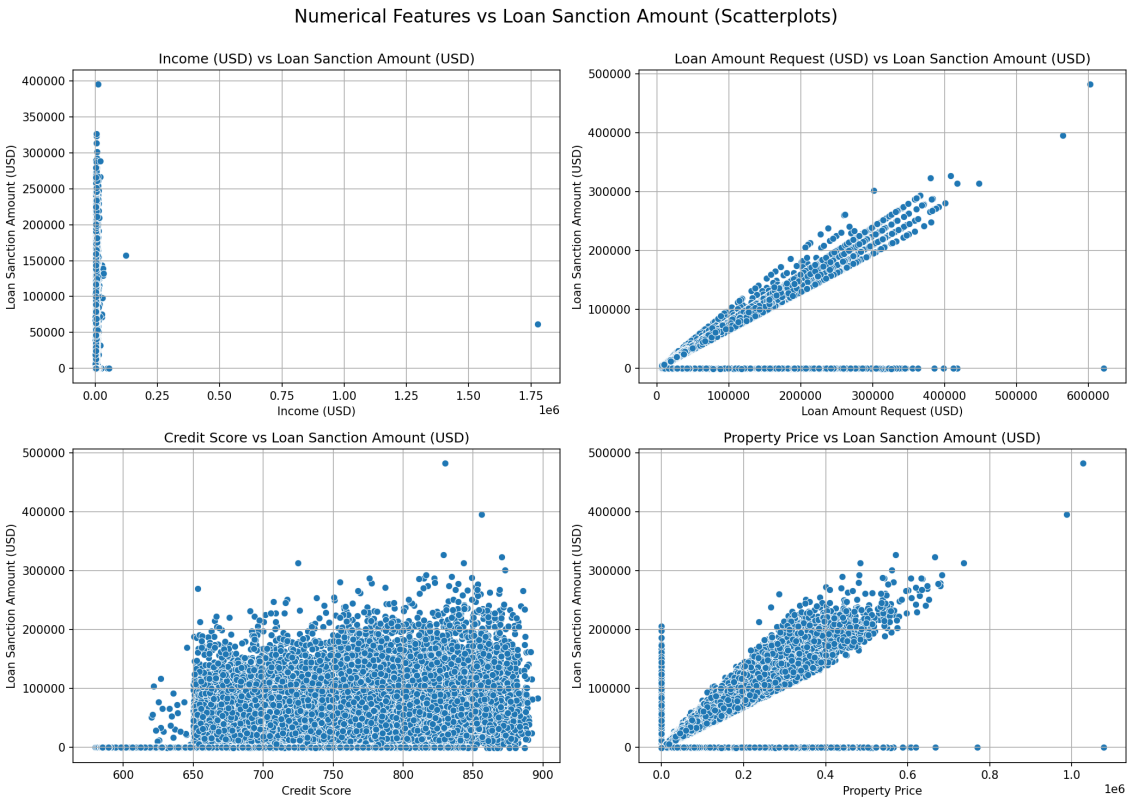


Figure 1: Scatter Plots for Numerical Features

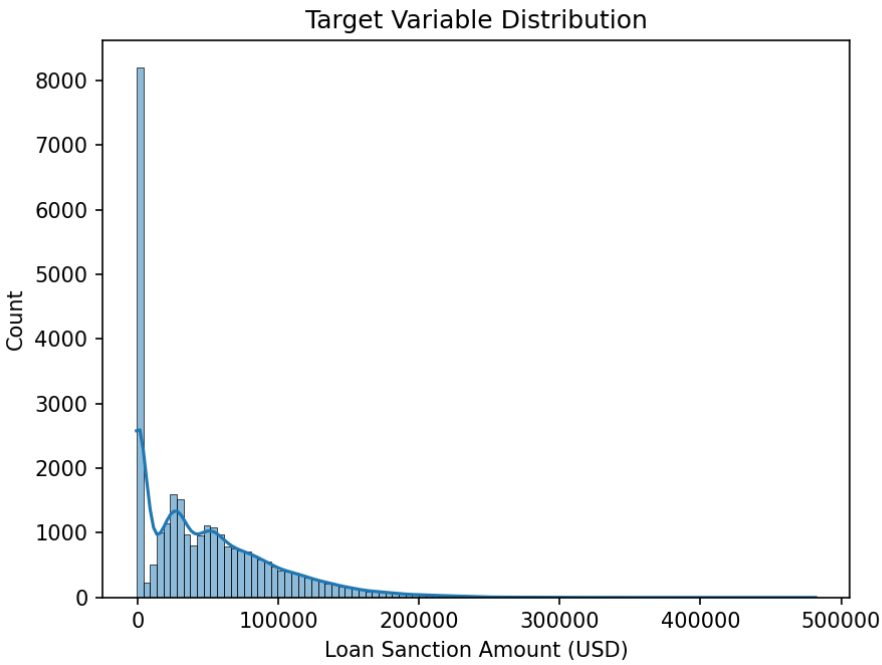


Figure 2: Target Variable Distribution

5.2 Linear Regression Results

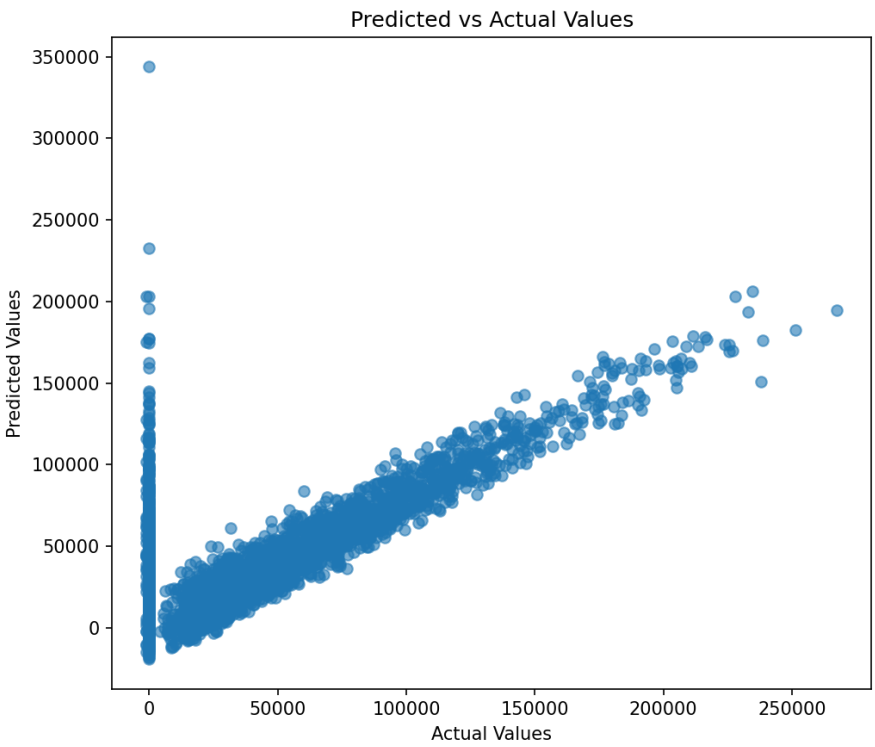
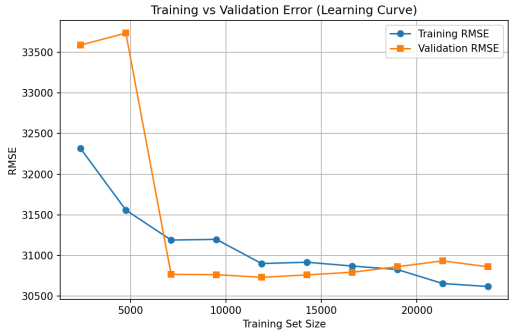


Figure 3: Predicted vs Actual (Linear Regression)



(a) Learning Curve

Figure 4: Linear Regression Model Plots

5.3 Lasso Regression Results

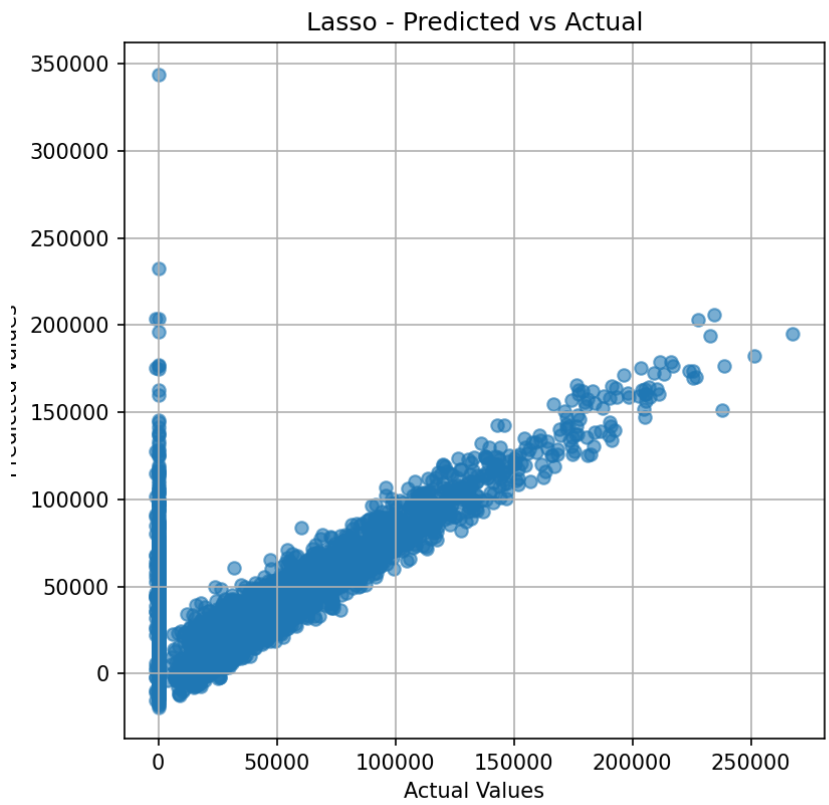
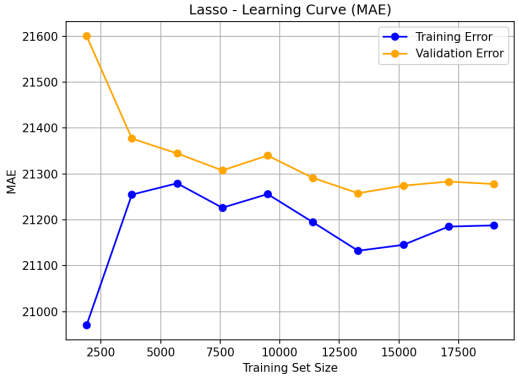


Figure 5: Predicted vs Actual (Lasso Regression)



(a) Learning Curve

Figure 6: Lasso Regression Model Plots

5.4 Ridge Regression Results

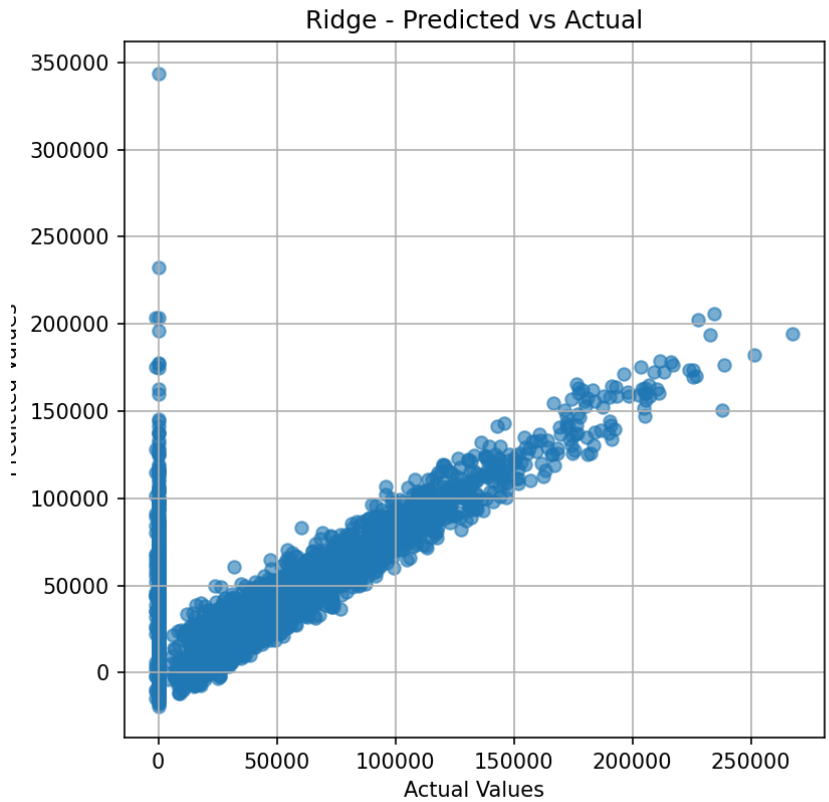
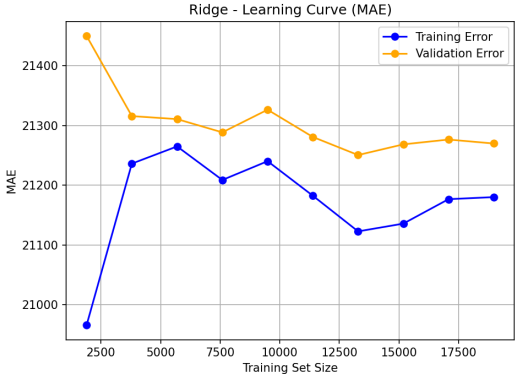


Figure 7: Predicted vs Actual (Ridge Regression)



(a) Learning Curve

Figure 8: Ridge Regression Model Plots

5.5 Elastic Net Regression Results

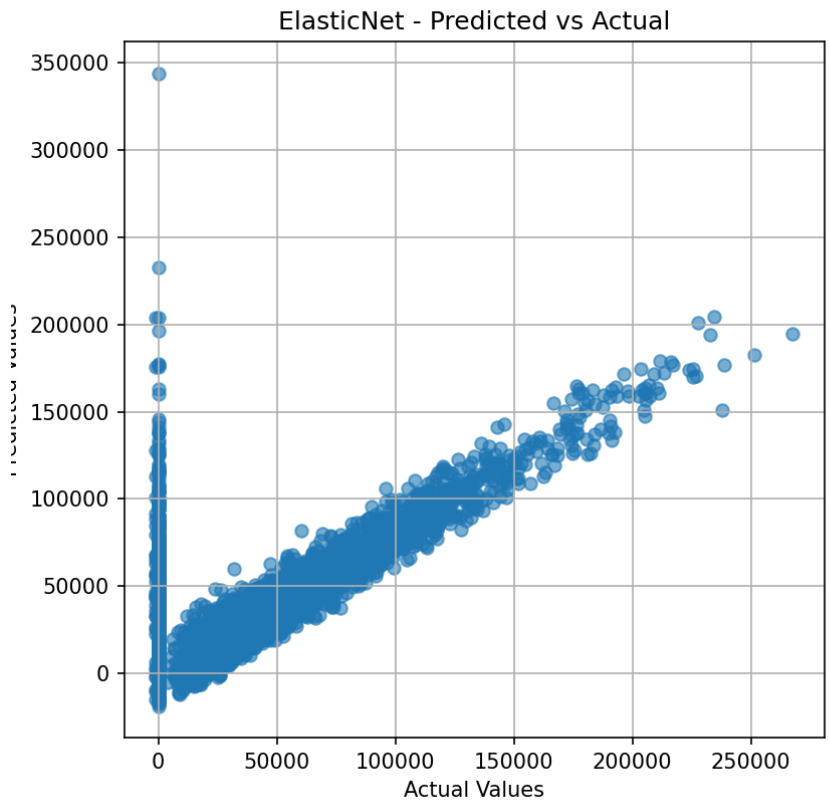
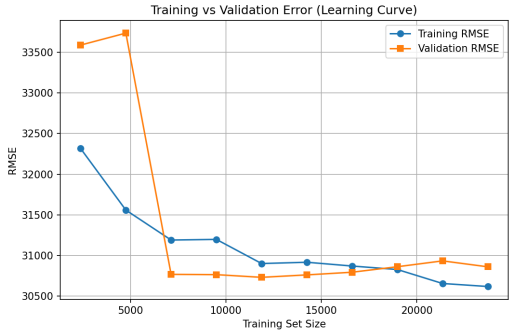


Figure 9: Predicted vs Actual (Elastic Net Regression)



(a) Learning Curve

Figure 10: Elastic Net Regression Model Plots

6 Performance Tables

6.1 Hyperparameter Tuning Results

Table 1: Hyperparameter Tuning Summary

Model	Search Method	Best Parameters	Best CV R^2
Ridge Regression	Grid / Random	$\alpha = 100$	0.5889
Lasso Regression	Grid / Random	$\alpha = 10$	0.5891
Elastic Net	Grid / Random	$\alpha = 0.01$, $l1 = 0.2$	0.5890

6.2 Cross-Validation Performance (K = 5)

Table 2: Cross-Validation Performance

Model	MAE	MSE	RMSE	R^2
Linear Regression	21103.11	1002429332.22	31661.16	0.5445
Ridge Regression	21086.35	1001589383.17	31647.90	0.5449
Lasso Regression	21089.94	1001445007.21	31645.62	0.5449
Elastic Net	21080.35	1001466603.48	31645.96	0.5449

6.3 Test Set Performance Comparison

Table 3: Test Set Performance

Model	MAE	MSE	RMSE	R^2
Linear Regression	21418.73	988649077.30	31442.79	0.5772
Ridge Regression	21424.25	988873190.77	31446.35	0.5771
Lasso Regression	21426.30	988395979.97	31438.77	0.5773
Elastic Net	21427.69	989141953.56	31450.63	0.5770

7 Overfitting and Underfitting Analysis

Training–validation error gaps indicate overfitting, while similar errors indicate good generalization. Regularization strength controls the bias–variance trade-off.

8 Bias–Variance Analysis

Ridge and Elastic Net reduce variance, Lasso promotes sparsity, and Linear Regression may underfit nonlinear patterns.

9 Observations and Conclusion

Regularized models improved generalization. Elastic Net provided the best balance between sparsity and stability.

References

- Scikit-learn Documentation: Linear Models
- Scikit-learn Documentation: Hyperparameter Optimization
- Loan Amount Dataset (Kaggle)